

UKSEDS COMPETITIONS

NATIONAL ROCKETRY CHAMPIONSHIP 2026-27

Requirement Specification

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MRC Midland Rocketry Club

Full Throttle Behaviour

Rockets & Things

WIZARD ROCKETS

UNITED KINGDOM ROCKETRY ASSOCIATION

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1. ACRONYMS & REFERENCE DOCUMENTS

1.1. Acronyms

Acronym	Description
NRC	National Rocketry Competition
RS	Requirement Specification
PDR	Preliminary Design Review
CDR	Critical Design Review
MR	Manufacturing Review
FRR	Flight Readiness Review
MRC	Midlands Rocketry Club
ECSS	European Space Agency's European Cooperation for Space Standardization
GEVS	NASA's General Environmental Verification Specification
ISO	International Standards Organization
RSO	Range Safety Officer
NOTAM	Notice To Airmen
UKRA	United Kingdom Rocketry Association
CAA	Civil Aviation Authority
COTS	Commercial Off The Shelf

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1.2. Reference Documents

1.2.1. Competition Documents

There are two other documents related to the competition, for a total of **three regulatory documents**. The three documents are as follows:

ID	Document	Description
NRC-RB	Rulebook	Outlines the overall competition rules, funding, team structures and so on - the competition logistics.
NRC-SoW	Statement of Work (This Document)	Outlines the main deliverables teams shall be expected to deliver - the competition project requirements.
NRC-RS	Requirements Specification	Outlines the technical requirements the rocket must be designed to meet

1.2.2. Useful Reference Documents

No.	Document	Description
1	ECSS-M-ST-10C Rev. 1	**Space project management** - Project planning and implementation
2	ECSS-S-ST-00-01C	**Glossary of Terms** - A list of commonly used terms in ECSS documents

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2. Introduction

2.1. Scope

This document outlines the requirements for the National Rocketry Championship (NRC). This document contains two parts: NRC-RS-TSN (Technical Specification) and NRC-RS-MRD (Mission Requirements Document). Both parts adhere to the **ECSS (European Cooperation for Space Standardization)**, in order to teach participating teams the level of precision, formality, and traceability employed in industry.

The Technical Specification (RS-TSN) outlines the technical design requirements that all teams' launch vehicles must comply with. These requirements are informed by Health & Safety, United Kingdom Rocketry Association (UKRA) rules and an obligation to maintain an acceptable baseline of engineering quality. Failure to meet these requirements may result in teams failing the Flight Readiness Review (FRR) and therefore being denied permission to launch.

The Mission Requirements Document (RS-MRD) is intended to guide participating teams towards a 'successful mission' on launch day. These requirements mirror practices in industry, where a "successful launch" differs from a "successful mission". This includes every requirement for the launch vehicle's flight, "The Customer's" payload and the rideshare payload. As mentioned in the statement of work (SOW), "The Customer's" payload will be provided by UKSEDS, but the rideshare payload should be provided by the participating team.

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2.2. Requirement Numbering Definition

The following describes the layout of the requirements within this document.

NRC RS-<TSN/MRD>-<A>-<n> | Rev: <R>

<Title>

<text>

VV Method: <*v*>
Parent: <*p*>

Where:
<RS-TSN> : Technical Specification
<RS-MRD> : Mission Requirements Document

<A>: Acronym for the section the requirement falls under.
<n>: Requirement number of 3 digits, starting from 001.
Note: <A> and <n> combine with the competition and document name acronyms to create the requirement ID.

<R>: Revision indicator, with "A" denoting the first version, "B" the second and so forth.
<Title>: Requirement title.
<text>: Requirement text.

<*v*>: the verification method(s) needed to demonstrate compliance to each requirement. The letters shall be only the following:
I: Inspection
D: Demonstration
A: Analysis
T: Test

<*p*>: Denotes if a requirement is derived from another requirement. Parent requirements shall be listed by the Requirement ID. If no Parent Requirement exists, the value shall be N/A.

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3. Motors [MOR]

3.1. Mounting of the Motor to the Launch Vehicle

NRC RS-TSN-MOR-001 | Rev: A

The launch vehicle motor/motor case shall be clamped axially at both ends.

The teams must provide a means of securely mounting the motor within the launch vehicle. There are two types of motors that comply with NRC's rules: single-use motors and reloads. Single-use motors do not require a case, but reloads do. Therefore, teams should clamp a single-use motor directly or clamp the case for a reload.

The mount must transmit the thrust loads from the motor to the structure of the launch vehicle, provide axial alignment of the motor within the launch vehicle and prevent the motor from slipping out during handling and all flight phases.

An example motor mount arrangement is shown in Figure RS-1. The motor can be clamped at the motor thrust ring to the base of the launch vehicle. Please remember this configuration is just an example and may not work for every team's purposes.

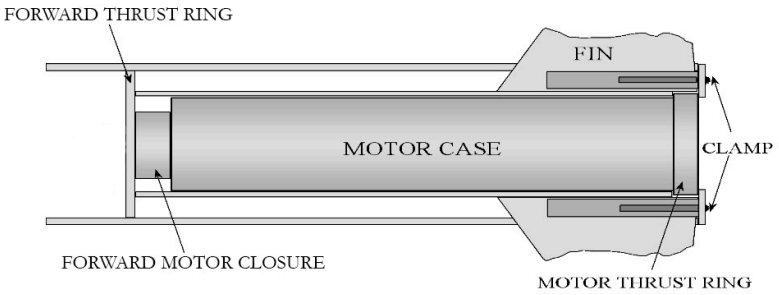


Figure RS-1: Motor mount arrangement schematic

VV Method: D
Parent: N/A

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3.2. Motor Mount Strength Requirements

NRC RS-TSN-MOR-002 | Rev: A

The launch vehicle motor mount shall be designed to withstand an axial force equivalent to twice the maximum motor thrust without permanent deformation.

VV Method: T
Parent: N/A

NRC RS-TSN-MOR-003 | Rev: A

The launch vehicle motor mount shall be designed to withstand a lateral force in any direction equivalent to the maximum motor motor thrust with a misalignment of 5° cantilevered about the vehicle's centre of gravity without permanent deformation (yield).

The motor mount must be designed to take all the thrust loads, both axial and lateral, from the motor thrust ring (referred to in RS-1). The motor mount must be designed to withstand an axial force equivalent to twice the maximum motor thrust; as well as a lateral force in any direction, equivalent to the maximum motor thrust with an axial misalignment of 5° cantilevered about the vehicle's centre of gravity. The motor mount must then transmit these loads into the body tube without permanent deformation (yield).

VV Method: T
Parent: N/A

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4. Launch Vehicle Stability Requirements [SMN]

All launch vehicles participating in the NRC must comply with certain stability criteria. Normally, stability will be verified using slender body theory. For slender body theory to apply, the following four constraints are set on the vehicle. This section outlines the minimum requirements to ensure a stable launch vehicle.

4.1. Length to Diameter Ratio (L/D)

NRC RS-TSN-SMN-001 | Rev: A

The launch vehicle shall have a minimum L/D of 15.

All rockets participating in the NRC must comply with certain stability criteria, normally verified using slender body theory.

VV Method: D
Parent: N/A

4.2. Normal Force Coefficient CN

NRC RS-TSN-SMN-002 | Rev: A

The launch vehicle shall have a normal force coefficient between 15 and 30.

All rockets participating in the NRC must comply with certain stability criteria, normally verified using slender body theory.

VV Method: A
Parent: N/A

4.3. Flight Velocity

Flight velocity is a critical factor in maintaining vehicle stability. This subsection defines the minimum safety requirements related to flight velocity that must be satisfied to ensure stable and safe vehicle operation throughout the flight.

NRC RS-TSN-SMN-003 | Rev: A

The launch vehicle shall have a maximum velocity of less than Mach 1 throughout all the phases of the flight.

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Mach 1 velocity is roughly equivalent to 343 m/s. All rockets participating in the NRC must comply with certain stability criteria, normally verified using slender body theory. Please note it is highly unlikely that your rocket will breach this, and if you have concerns you may wish to contact UKSEDS for clarification.

VV Method: A
Parent: N/A

NRC RS-TSN-SMN-004 | Rev: A

The launch vehicle shall have a minimum velocity of 20 m/s when leaving the launch rail.

At Midlands Rocketry Club NRC launch vehicles usually fly with 2m rails.

VV Method: A
Parent: N/A

4.4. Stability Margin

NRC RS-TSN-SMN-005 | Rev: A

The launch vehicle shall have a static stability margin between 1.75 diameters and 2.75 diameters during all phases of flight before apogee.

The static stability margin is determined by the relative positions of the centre of mass and the centre of pressure (see Figure RS-2). It represents the stability of the launch vehicle under static conditions, before aerodynamic and flight dynamics effects are considered.

Dynamic stability describes how the launch vehicle's stability changes during flight. This is typically assessed by evaluating the stability margin throughout the trajectory and may be presented as a plot of stability margin against altitude. These plots are crucial to the evaluation of your vehicle's design and flight readiness. Failure to provide sufficient evidence of dynamic stability, including the relevant stability-margin plots, may result in the vehicle not being approved for launch.

If the basic criteria above cannot be met, further documentation must be presented to the RSO to demonstrate that the launch vehicle is both statically and dynamically stable.

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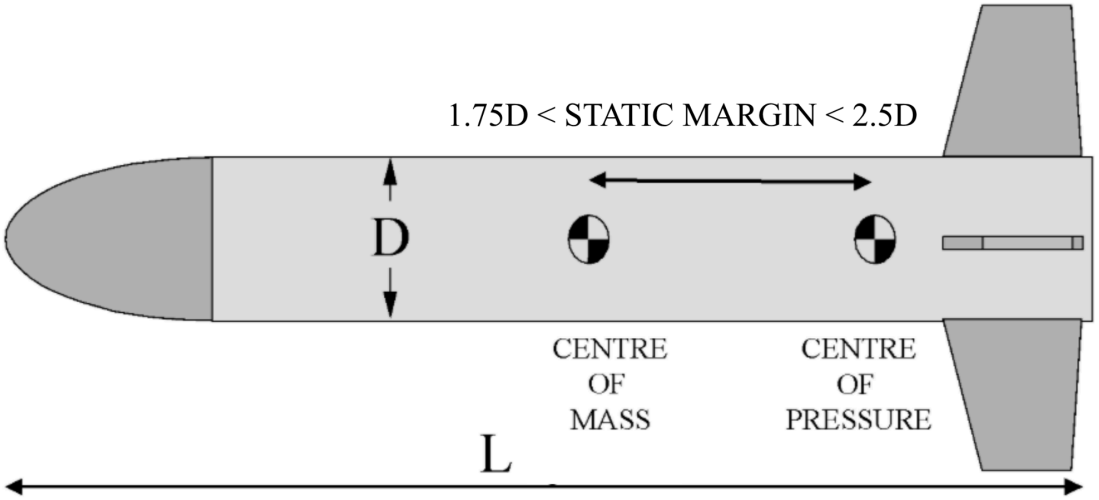


Figure RS-2: Schematic to illustrate the stability margin on a rocket

VV Method: A
Parent: N/A

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5. Structural Acceptance Requirements [SAR]

For a launch vehicle to be accepted for flight it must pass certain pre-launch criteria, which include structural tests. Passing these tests will not guarantee that a launch vehicle will have a flawless flight but will ensure a minimum chance of failure and permit the launch vehicle to be safe for flight.

5.1. Fin Alignment

NRC RS-TSN-SAR-001 | Rev: A

The geometric alignment of each fin shall be within 2° of the projected longitudinal axes of symmetry of the launch vehicle.

See Figure RS-3 for more information.

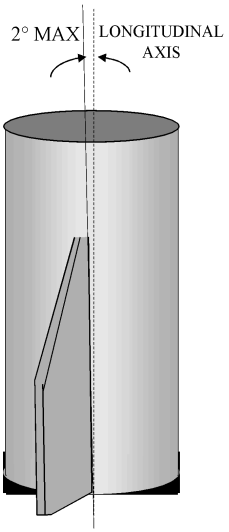


Figure RS-3: Schematic of to illustrate geometric alignment constraint of fins

VV Method: I

Parent: N/A

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5.2. Fin Longitudinal Loading

NRC RS-TSN-SAR-002 | Rev: A

Each fin must be able to support a suspended load from its tip equivalent to twice the fin mass times the launch vehicle's maximum axial acceleration occurring during any flight phase without permanent deformation.

Please see **Figure RS-4** for more details.

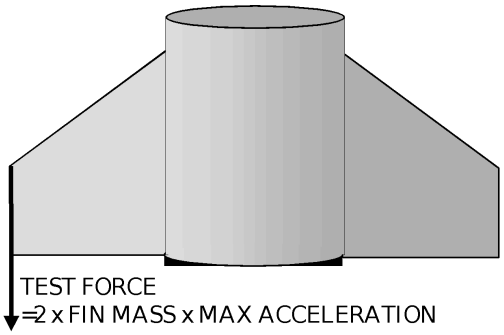


Figure RS-4: Schematic to illustrate of longitudinal loading force on individual fin

VV Method: T
Parent: N/A

5.3. Fin Lateral Loading

NRC RS-TSN-SAR-003 | Rev: A

Each fin shall withstand a transverse load in either direction, equivalent to the launch vehicle's launch mass when suspended from the fin tip with a maximum lateral deflection of less than 10°.

See Figure RS-5 for more information.

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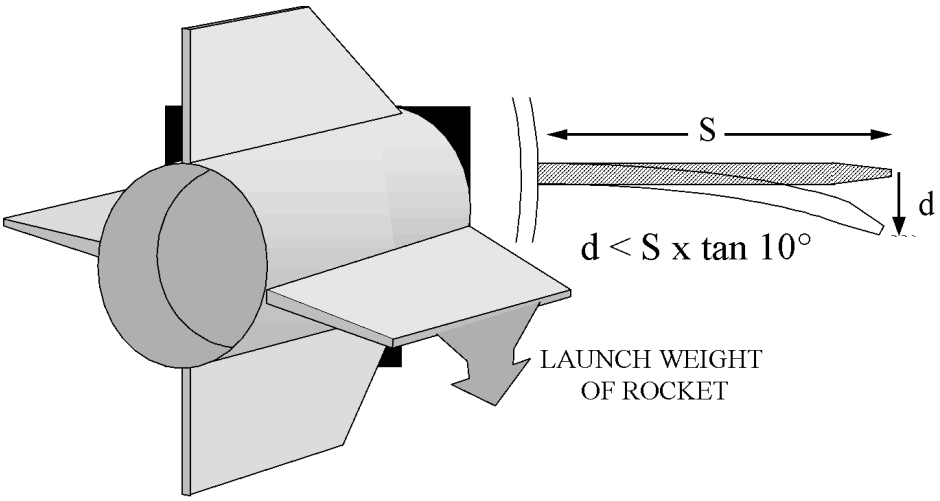


Figure RS-5: Schematic to illustrate of lateral loading force on individual fin

VV Method: T
Parent: N/A

5.4. Fuselage Stiffness

NRC RS-TSN-SAR-004 | Rev: A

The launch-ready launch vehicle shall deflect less than 10mm per metre length in any lateral direction when suspended from its centre of mass.

See Figure RS-6 for more information.

VV Method: T
Parent: N/A

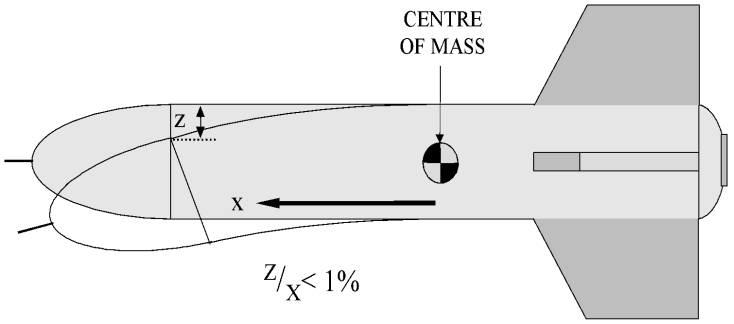


Figure RS-6: Schematic to illustrate fuselage stiffness of rocket

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NRC RS-TSN-SAR-005 | Rev: A

The mating length between a coupler and body tube shall be a minimum of 1 body tube diameter.

When a **launch vehicle** is coupled together from several tubes, the method of joining the tubes is left to the application and discretion of the designer. However, it is required that the mating length between the coupler and each tube should be a minimum of 1 diameter when using plastics or composites (see Figure RS-7). This is advised to maintain satisfactory levels of stiffness along the length of the **launch vehicle**.

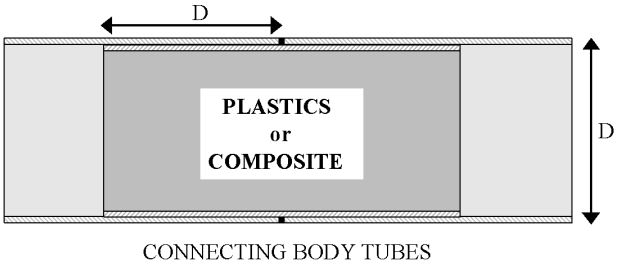


Figure RS-7: Schematic to illustrate mating length between coupler and body tube

For sliding connections, the minimum mating length between the coupler and tube should always be 1 diameter. Additionally, the fit between parts must stop any noticeable rotation at the joint.

VV Method: D
Parent: N/A

5.5. Static Strength Requirements

NRC RS-TSN-SAR-006 | Rev: A

All structural parts shall withstand twice their expected maximum inertial and aerodynamic loads without failure.

If participants wish to re-use their vehicles, all parts must withstand 150% of the applied loads without permanent deformation (yield).

VV Method: T
Parent: N/A

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5.6. Launch Lug

NRC RS-TSN-SAR-007 | Rev: A

All launch vehicles shall be equipped with a minimum of 2 rail buttons.

At Midlands Rocketry Club we usually fly on 2m rails which are compatible with 6mm launch buttons. We recommend using Delrin rail buttons and an example can be found here. A recommended layout of the launch buttons can be found in Figure RS-8.

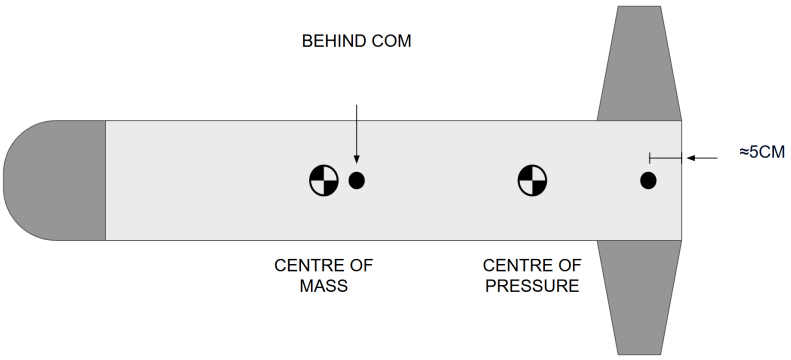


Figure RS-8: Schematic to illustrate recommended layout for launch lugs

VV Method: D

Parent: N/A

5.7. Finite Element Analysis

NRC RS-TSN-SAR-008 | Rev: A

Finite Element Analysis shall not be used to produce evidence for any structural requirements.

Whilst we would encourage teams to learn finite element analysis (FEA) for NRC and welcome its use in competition reports and documentation as supplementary evidence, for safety reasons, we cannot accept FEA evidence for verifying any structural requirements. We have seen teams in previous editions of NRC making incorrect assumptions, for example assuming homogenous material properties for additive manufacturing methods or 3D printing. These assumptions have led to issues at the flight readiness review where they find their design does not pass real-world structural tests. FEA is better suited to assessing trade-offs for material or design choices prior to physical testing occurring.

VV Method: D

Parent: N/A

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6. Electrical Systems [ESS]

Teams may include several electrical subsystems onboard their launch vehicles, such as payloads, a telemetry system and an altimeter.

6.1. Internal Power Supply

NRC RS-TSN-ESS-001 | Rev: A

The launch vehicle shall be capable of being powered on and left to run using its internal power source for a minimum of 15 minutes on the launch pad.

Ideally, teams should have a way to arm the launch vehicle externally so that it can be armed on the launch pad. If this is not possible with the launch vehicle design chosen then the team will need to power it for much longer (up to an hour) on the competition day before it is placed on the pad. This should be budgeted for in addition to the energy used during the predicted total flight time.

VV Method: A
Parent: N/A

NRC RS-TSN-ESS-002 | Rev: A

The launch vehicle shall be capable of being armed at the launch site within 5 minutes.

Many delays on competition days are caused by teams having to fully disassemble their launch vehicles to arm them or switch on a payload. Teams must ensure that their launch vehicle/payload is externally armable. This can be done with pull pins or screw switches for example.

VV Method: D
Parent: N/A

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6.2. Electrical Considerations

NRC RS-TSN-ESS-003 | Rev: A

The launch vehicle shall not contain single-core wire.

Single-core wire shall not be used for electrical cable in primary systems since it is considered too fragile. Multi-core[1] wire is tougher and more reliable under handling and flight conditions so this shall be used instead.

During assembly, testing and flight, electronic circuits and wiring are subjected to a high level of abuse. This requires that launch vehicle construction should be rugged, tidy and of a high standard of workmanship as possible.

Care should be taken when using electromechanical components (such as relays, switches, and connectors) to ensure that they are capable of withstanding high acceleration and vibration loads. Care should be taken to ensure that the switch is placed in the correct orientation. This is critical as they can only withstand high G loads along certain axes. Always refer to the manufacturer's data sheets.

The circuit must be structurally and electrically robust so that no parts of the circuit can change state or function due to any mechanical loads from transportation, manipulation on the launch pad or in flight.

Recovery system deployment shock loads are a very important design case and can exceed the thrust loads.

VV Method: D
Parent: N/A

[1] Also known as multi-stranded wire

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6.3. Radio Frequencies

NRC RS-TSN-ESS-004 | Rev: A

Teams shall conform with Ofcom's IR 2030 radio frequency regulations.

A copy of Ofcom's IR 2030 can be found here. All teams must ensure that they comply with these regulations when using radios.

Generally, teams are caught out with 915MHz which they will find is not permitted for airborne use. Depending on the specifics of the system, 868MHz and 433MHz are the most popular choices but please make sure to check the latest IR 2030 document.

VV Method: D
Parent: N/A

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7. Recovery Requirements [RRS]

7.1. Use of Recovery System

NRC RS-TSN-RRS-001 | Rev: A

The launch vehicle shall have a COTS or UKRA qualified recovery system.

To minimize health risk and hazards, all launch vehicles must have a Commercial Off The Shelf (COTS) **or UKRA qualified** system to recover them in a safe and controlled manner. This usually takes the form of a parachute that is activated when the launch vehicle reaches apogee.

VV Method: D
Parent: N/A

7.2. Landing Speeds

NRC RS-TSN-RRS-002 | Rev: A

The recovery system shall reduce the launch vehicle's vertical landing speed to less than 15 m/s.

This can be shown through OpenRocket simulations if teams use a parachute based recovery system. If teams use an alternative recovery system then they shall demonstrate this through hand calculations and possibly real-world tests.

VV Method: A
Parent: N/A

7.3. Maximum Post-Apogee Range

NRC RS-TSN-RRS-003 | Rev: A

After apogee, the launch vehicle shall not drift more than one kilometre before landing at wind speeds of 15m/s.

The prospective launch site is quite windy, but all launch vehicles must still land within the landowner's boundaries. It is advisable to make the recovery device highly visible to assist tracking.

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VV Method: A
Parent: N/A

7.4. Flight Computer

NRC RS-TSN-RRS-004 | Rev: A

Any flight computer used for recovery shall be a COTS or UKRA qualified system.

The purpose of a flight computer is to activate the recovery system when the launch vehicle reaches apogee.

An alternative method to using the motor's ejection charge is to use an electronic timer. This timer is activated on launch and is pre-programmed to wait for a set time interval before firing the recovery system. Some devices can be triggered by achieving a given acceleration, velocity or altitude. Essentially, this requirement says that teams cannot make their own flight computer and use it to trigger the recovery system, they must use a commercially available one. They may, however, make a flight computer and fly it as a payload, as long as it is not connected to the recovery system.

There is an exception to this requirement which is a SRAD (student researched and developed) or homemade flight computer that has been flight-qualified to UKRA's standards. If teams wish to develop their own recovery circuit, as per the UKRA safety code section 2, this will be classed as an experimental launch vehicle and so UKRA section 8 applies. This means that the **"homebrew" recovery electronics must be flown as a backup to a COTS recovery system** until they have been qualified as per section 8 of the UKRA Safety Code.

The UKRA Safety Code can be found here.

VV Method: D
Parent: N/A

7.5. Isolation of Recovery Circuit

NRC RS-TSN-RRS-005 | Rev: A

Electronic recovery circuits shall be electrically isolated from other electronic systems.

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It is recommended that all of the recovery sequencing circuitry (including the battery) be electrically isolated from any other electrical circuit used in the launch vehicle.

VV Method: D
Parent: N/A

7.6. Detection of Apogee

NRC RS-TSN-RRS-006 | Rev: A

Tilt switches shall not be used.

Detectors relying on the physical orientation of the launch vehicle relative to the gravity vector to detect apogee (such as tilt switches) are inaccurate and should not be used.

VV Method: D
Parent: N/A

7.7. Flight Computer Disarming Mechanism

NRC RS-TSN-RRS-007 | Rev: A

Any flight computers used to trigger recovery systems shall have an externally accessible disarming mechanism.

The flight computer (if used) must have a safe and secure disarming mechanism, which prevents inadvertent activation of the recovery system during handling and loading. This is especially important where pyrotechnic actuators are used.

The recovery system must be kept in the disarmed (safe) condition until the launch vehicle is safely loaded into the launch pad. At the designated point during countdown, the launch vehicle can then be armed.

VV Method: D
Parent: N/A

7.8. Transmission of Recovery Shock Loads

NRC RS-TSN-RRS-008 | Rev: A

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Recovery loads shall be transmitted via eye bolts through bulkheads with a nut on the other side.

The main recovery shock loads must not be transmitted in shear through screw threads into the launch vehicle body as shown in Figure RS-9. It is recommended that these loads be transmitted through links and hook-eye anchor points.

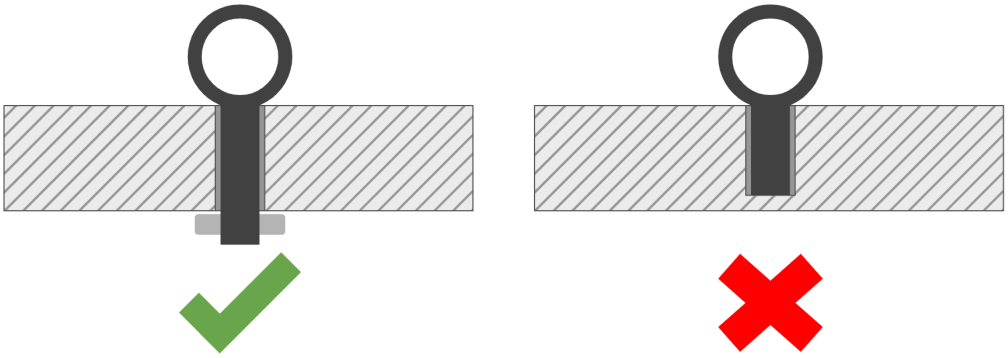


Figure RS-9: Schematic to illustrate the transmitted loads through bulkhead

VV Method: D
Parent: N/A

7.9. Shock Cord

NRC RS-TSN-RRS-009 | Rev: A

Recovery systems incorporating shock cords shall use a shock cord with a minimum length equal to five times the total length of the launch vehicle.

A lot of energy is released when the ejection charge fires. This energy must be great enough to separate the airframe and so if the shock cord is very short then there may not be sufficient cord length to dissipate this energy, leading to the shock cord snapping.

VV Method: D
Parent: N/A

NRC RS-TSN-RRS-010 | Rev: A

Shock cords shall be made of Kevlar.

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Previous NRC editions, as well as wider rocketry community flying, suggest that this is the best material to use for shock chords. It is also relatively inexpensive compared to alternative materials.

VV Method: D
Parent: NRC RS-TSN-RRS-007

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8. Alternative Recovery [ARY]

8.1. Alternative Recovery Definition

NRC RS-TSN-ARY-001 | Rev: A

All launch vehicles shall conform to CAA UAS laws.

This includes the regulation of any payload deployment.

VV Method: D
Parent: N/A

NRC RS-TSN-ARY-002 | Rev: A

All launch vehicles shall conform with the UKRA Safety Code.

The UKRA Safety Code can be found here.

VV Method: D
Parent: N/A

UKSEDS define alternative recovery methods as **any** method used to slow a launch vehicle down that does not use a conventional parachute.

The Civil Aviation Authority (CAA) and United Kingdom Rocketry Association (UKRA) have very strict laws and rules governing what can and cannot be done with rockets. To help teams adhere to them, the organisers have added the below sections as guidance.

Disclaimer: Please always refer to the CAA and UKRA's websites for up-to-date guidance and information. If you find any discrepancies between this document and their websites, please follow the CAA and UKRA's guidance and let the organisers know at: rocketry@ukseds.org

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8.2. Alternative Deployment Methods

For any deployment that occurs **above** the CAA's drone limit of 120m/400ft either:

1. The ejection charge within the motor
2. A COTS flight computer

must be used to trigger any deployment.

For the deployment of any Unmanned Aircraft Systems (UAS) from a launch vehicle, this **must** occur **below** the CAA's drone limit of 120m (400ft). The maximum distance a UAS should be flown from its operator is 500 metres or not beyond the visible line of sight if less than 500 metres.[2]

More information about the CAA's drone laws can be found here.

However, for safety reasons, the deployment of the launch vehicle's first stage recovery system must be triggered within 150m of apogee and the final stage (or the first stage if single stage) must deploy no lower than an altitude of 200m.

Any alternative recovery system designs proposed by a team in their reports will be submitted to the UKRA Safety and Technical Committee for approval before launch.

8.3. Altitude and Deployment Limits for UKRA Clubs

A list of active UKRA clubs can be found here.

Please note that no person or team can break the maximum altitude of any site. This is set by the CAA and cannot be altered.

If your team is going to go above the dual deployment limit of any club, you must use a dual deployment system. For more information on dual deployment, please refer to Apogee Rocket's guide here

[2] It is the understanding of UKSEDS that any launch vehicle carrying a deployable UAS must not breach this 400ft drone limit. If your team intends carry any form of deployable payload please contact rocketry@ukseds.org

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8.4. Active Stabilisation

Active stabilisation systems may include steerable fins, airbrakes, canards, and other similar mechanisms. Teams may wish to incorporate such systems as part of their rideshare payload. However, the use of active stabilisation systems is subject to strict requirements to ensure compliance with applicable legislation and UKRA guidelines.

NRC RS-TSN-ARY-003 | Rev: A

Active stabilisation shall only be used on ascent.

As per the UKRA's safety code, active stabilisation can only be flown on ascent. This means that section 8 of the UKRA safety code **must** be followed.

The UKRA's safety code can be found here.

VV Method: D
Parent: N/A

NRC RS-TSN-ARY-004 | Rev: A

Active stabilisation systems shall only be used to maintain the launch vehicle's intended upward flight orientation.

The implemented active stabilisation system is intended to keep the launch vehicle oriented in the upward direction during flight and to correct deviations from its nominal course. It is not intended to deliberately steer the vehicle away from its planned trajectory. Using active stabilisation for intentional trajectory control may introduce legal and regulatory concerns and could affect whether the vehicle is permitted to fly.

VV Method: D
Parent: N/A

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NRC RS-TSN-ARY-005 | Rev: A

All active stabilisation systems shall incorporate a lockout mechanism.

The purpose of the lockout mechanism is to allow the launch vehicle to fly safely even if development or validation of the active stabilisation system has not been completed before launch day. UKSEDS encourages teams to take on ambitious engineering challenges and explore innovative solutions. At the same time, we want teams to retain the best possible chance of flying their vehicle. By incorporating a lockout mechanism, an incomplete or unvalidated active stabilisation system can remain disabled during flight, allowing the vehicle to be assessed and flown without relying on that subsystem.

VV Method: D
Parent: N/A

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9. Mission Requirements [MIS]

NRC RS-MRD-MIS-001 | Rev: A

The launch vehicle should reach an apogee of 2000ft.

Below is a table outlining the mark distribution with respect to altitude.

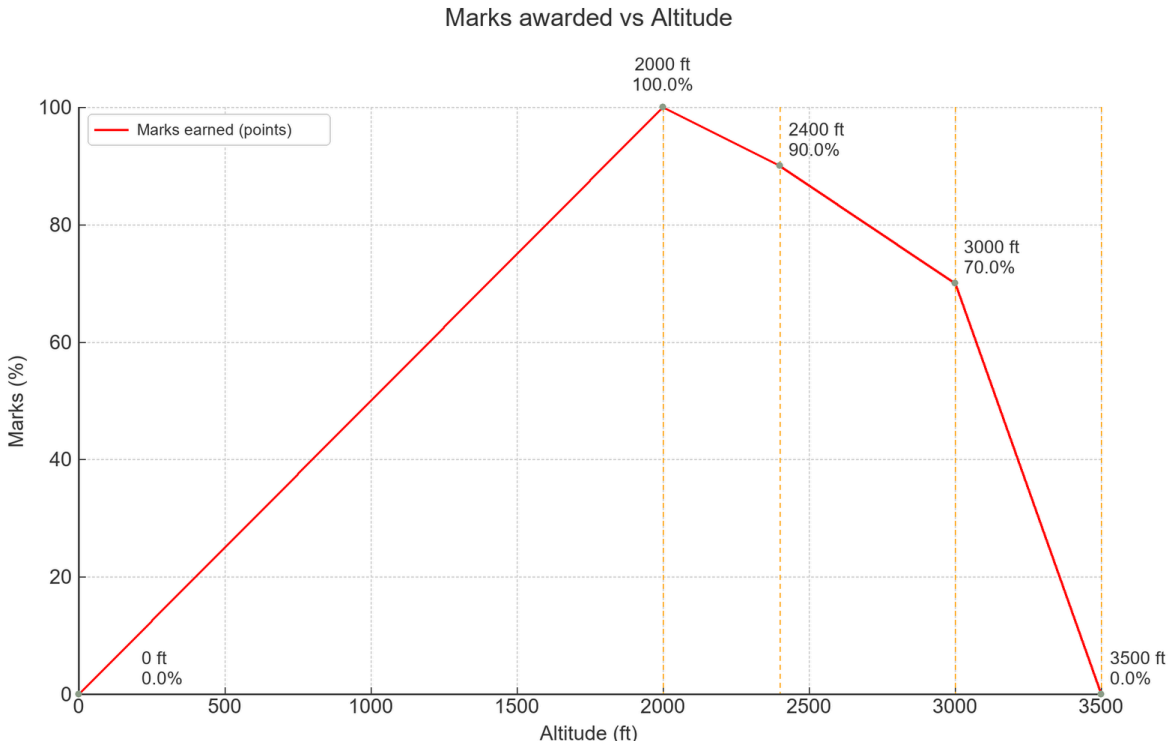


Figure RS-10 - Apogee vs Marks Awarded

VV Method: A

Parent: N/A

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NRC RS-MRD-MIS-002 | Rev: A

The launch vehicle shall not fly above 3000ft without a dual-deployment recovery system.

Rockets achieving an apogee above 3000ft are required to incorporate a dual deployment recovery system onboard as per Midlands Rocketry Club rules. Rockets designed by teams for NRC are *not required* to incorporate dual deployment, due to the target apogee being 2200ft. As such, any rocket exceeding a 3000ft apogee will be severely penalised.

VV Method: A

Parent: NRC RS-MRD-MIS-001

NRC RS-MRD-MIS-003 | Rev: A

The launch vehicle shall not fly above the Midlands Rocketry Club's NOTAM limit.

NOTAM stands for Notice To Airmen. NRC's launch site is overseen by the Midlands Rocketry Club, which can only secure a maximum altitude of 3500ft for the NRC competition day. Rockets are NOT permitted to fly above this altitude. This is a hard-and-fast rule. As such, any rockets breaching this maximum altitude will automatically receive 0 marks on the launch day.

VV Method: A

Parent: NRC RS-MRD-MIS-001

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NRC RS-MRD-MIS-004 | Rev: <A>

The launch vehicle shall be powered by a 2-grain 29mm G class ammonium perchlorate solid rocket motor with a total impulse under 130Ns.

Teams are limited to 2-grain 29mm Cesaroni reloads of G impulse and 29mm Aerotech G class motors with an impulse under 130Ns. These are classified as mid-power motors and so do not require UKRA certification to purchase or fly. **All motors must be ordered through UKSEDS**, the process for which will be communicated prior to the PDR submission. This mitigates communication and lead time issues as well as ensuring motors are handled correctly only by qualified and licensed individuals.

In the 2026/27 competition, UKSEDS are expecting single-use Aerotech motors, as well as single- and reloadable Cesaroni motors to be available. UKSEDS will endeavour to make a list of orderable motors available at the earliest opportunity. Please contact rocketry@ukseds.org regarding questions on motors, and do not directly contact suppliers, as this creates confusion.

VV Method: D

Parent: N/A

NRC RS-MRD-MIS-005 | Rev: A

The rocket motor used shall be manufactured by Cesaroni Technology Incorporated or AeroTech.

Due to safety and competition restrictions, only approved motors can be used in the NRC. UKSEDS will not allow self-made motors or motors from unlicensed manufacturers, as these are illegal. Motors from Cesaroni Technology Incorporated or AeroTech are UKSEDS verified and approved, and as such are the chosen suppliers for motors.

VV Method: D

Parent: NRC MRD-MIS-004

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NRC RS-MRD-MIS-006 | Rev: A

The rocket motor shall be obtained through UKSEDS' NRC organising team.

In previous NRC competitions we have had significant issues with communication between teams and the motor supplier. For this NRC edition, all teams shall order through UKSEDS so that the UKSEDS organising team is responsible for all communications with motor suppliers regarding motor orders for the competition day. This also ensures all of the motors are handled correctly by qualified persons with the correct licences.

VV Method: D

Parent: NRC RS-MRD-MIS-004

NRC RS-MRD-MIS-007 | Rev: A

The launch vehicle shall carry a rideshare payload with a minimum mass of 100g.

The rideshare payload shall be provided by the teams. The nature of this rideshare payload shall be determined by the team.

HINT: some of the other requirements might indicate some features that your payload should incorporate but you may add more!

VV Method: D

Parent: N/A

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10. "The Customer's" Payload [CPD]

Note: "The Customer's" payload will be provided on the day by UKSEDS. You are **not** expected to design or provide this. "The Customer's" payload has a power rating of 7.5W and a mass of 75g (not including the mass of the launch vehicle adapter).

NRC RS-MRD-CPD-001 | Rev: A

The launch vehicle shall provide a DC bus voltage of 3.5-5V to "The Customer's" payload.

VV Method: D,T

Parent: N/A

NRC RS-MRD-CPD-002 | Rev: A

The launch vehicle shall provide power to "The Customer's" payload for a minimum of 45 minutes.

This includes time on the launch pad, flight, and recovery. This requirement assumes that the launch vehicle is on the launch pad for 15 minutes.

VV Method: D

Parent: NRC RS-TSN-ESS-001

NRC RS-MRD-CPD-003 | Rev: A

The launch vehicle shall provide sufficient current to power "The Customer's" payload.

VV Method: D,T

Parent: N/A

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NRC RS-MRD-CPD-004 | Rev: A

The launch vehicle shall provide power via a female XT30 connector at the top and bottom of the "The Customer's" payload.

Please note: Either the top or bottom port can be used to provide power to the payload. Two options have been provided to give participants more flexibility in their designs.

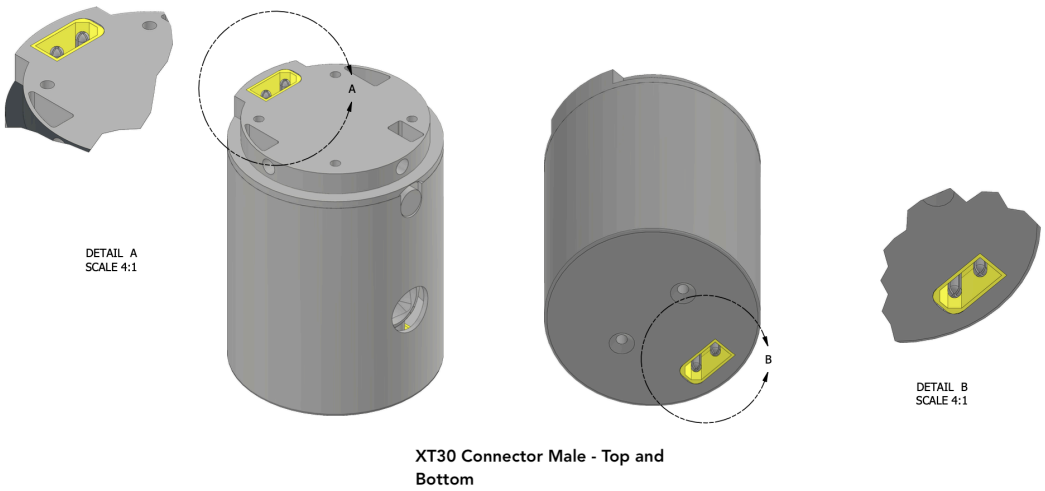


Figure RS-11 :"The Customer's" payload showing XT30 connector locations top and bottom



Figure RS-12 Male XT30 connector (left) to be provided by UKSEDS. Female XT30 connector (right) to be provided by participants.

VV Method: D

Parent: N/A

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NRC RS-MRD-CPD-005 | Rev: A

The launch service provider shall provide a microSD with a minimum of 16GB storage for "The Customer's" payload.

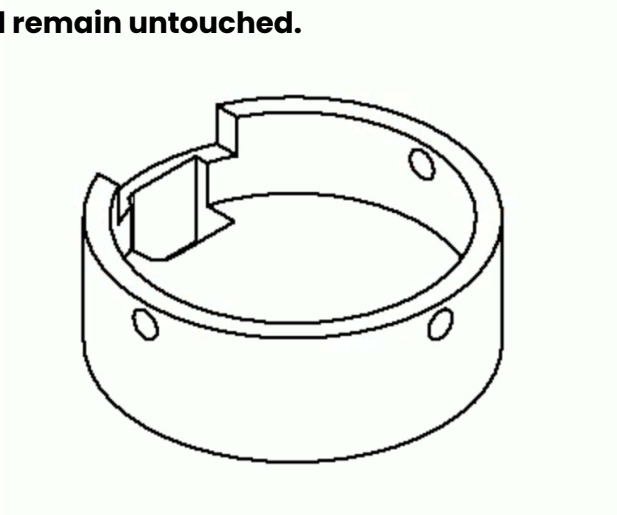
VV Method: D

Parent: N/A

NRC RS-MRD-CPD-006 | Rev: A

The launch vehicle shall mechanically interface with "The Customer's" payload via the launch vehicle adapter.

The launch vehicle adapter CAD file has been shared with teams through the documentation page on the website[3]. **Participating teams** are expected to manufacture and integrate the launch vehicle adapter in their designs. This has been done so that you design for and include the correct adapter in your design. It should also reduce interface issues on the competition day. **Teams are welcome to modify the launch vehicle adapter to interface with their designs. However the face that interfaces with the payload as well as the internal volumes and dimension of the 'inner walls' should remain untouched.**



Launch vehicle adapter

Figure RS-13 : Launch Vehicle adapter

[3] UKSEDS will communicate any alterations which may occur to the adapter.

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VV Method: D, T

Parent: N/A

NRC RS-MRD-CPD-007 | Rev: A

The launch vehicle shall provide a minimum payload bay of 40mm diameter and 53mm length for "The Customer's" payload (not including the launch vehicle adapter). Refer to Appendix A

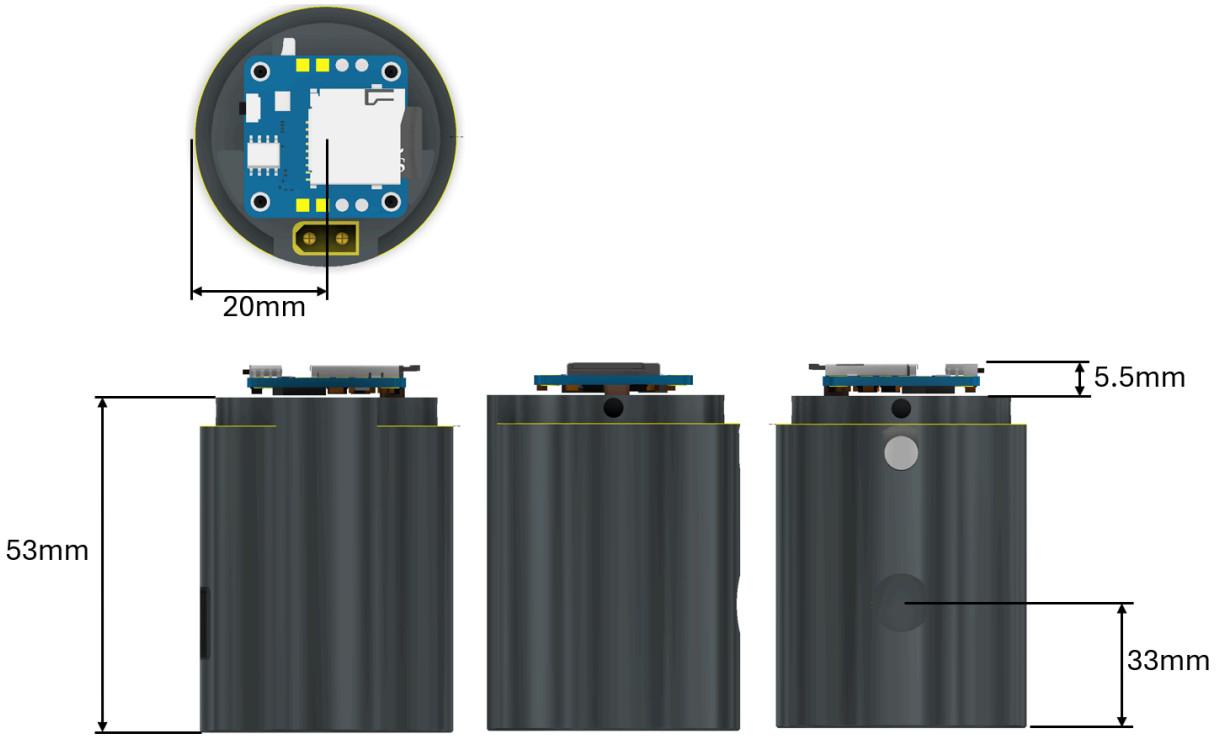


Figure RS-14 - Customer's Payload

Important Note : Teams are advised to design their payload bays to have a small amount of extra space to account for any tolerancing issues at the integration stage. It is highly recommended to print out a copy of the dummy payload and carry out fit checks and work out an assembly procedure for the competition day.

VV Method: D

Parent: N/A

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NRC RS-MRD-CPD-008 | Rev: A

The thrust vector shall go through the geometric centre of “The Customer’s” payload.

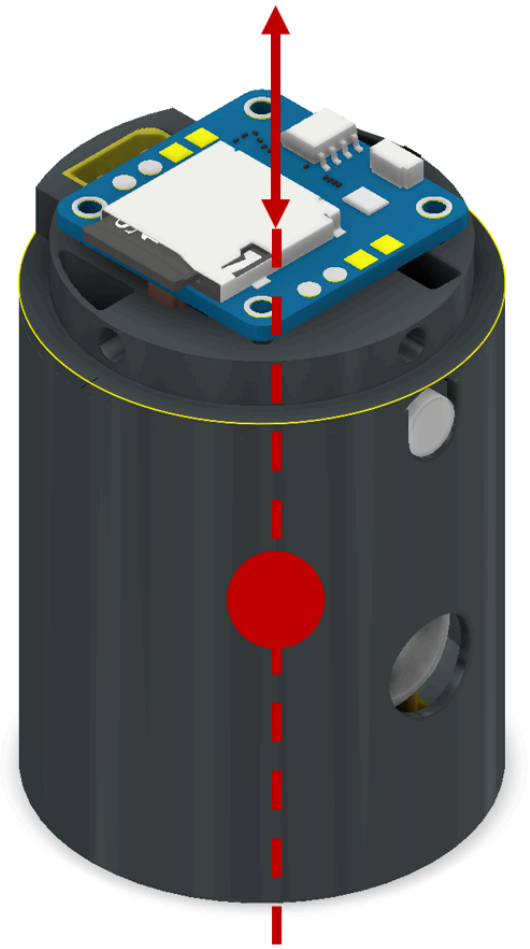


Figure RS-15 : Central Axis

VV Method: D

Parent: N/A

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NRC RS-MRD-CPD-009 | Rev: A

The launch vehicle shall provide permanent external access to the following points for "The Customer's" payload (circled red). Refer to Appendix A

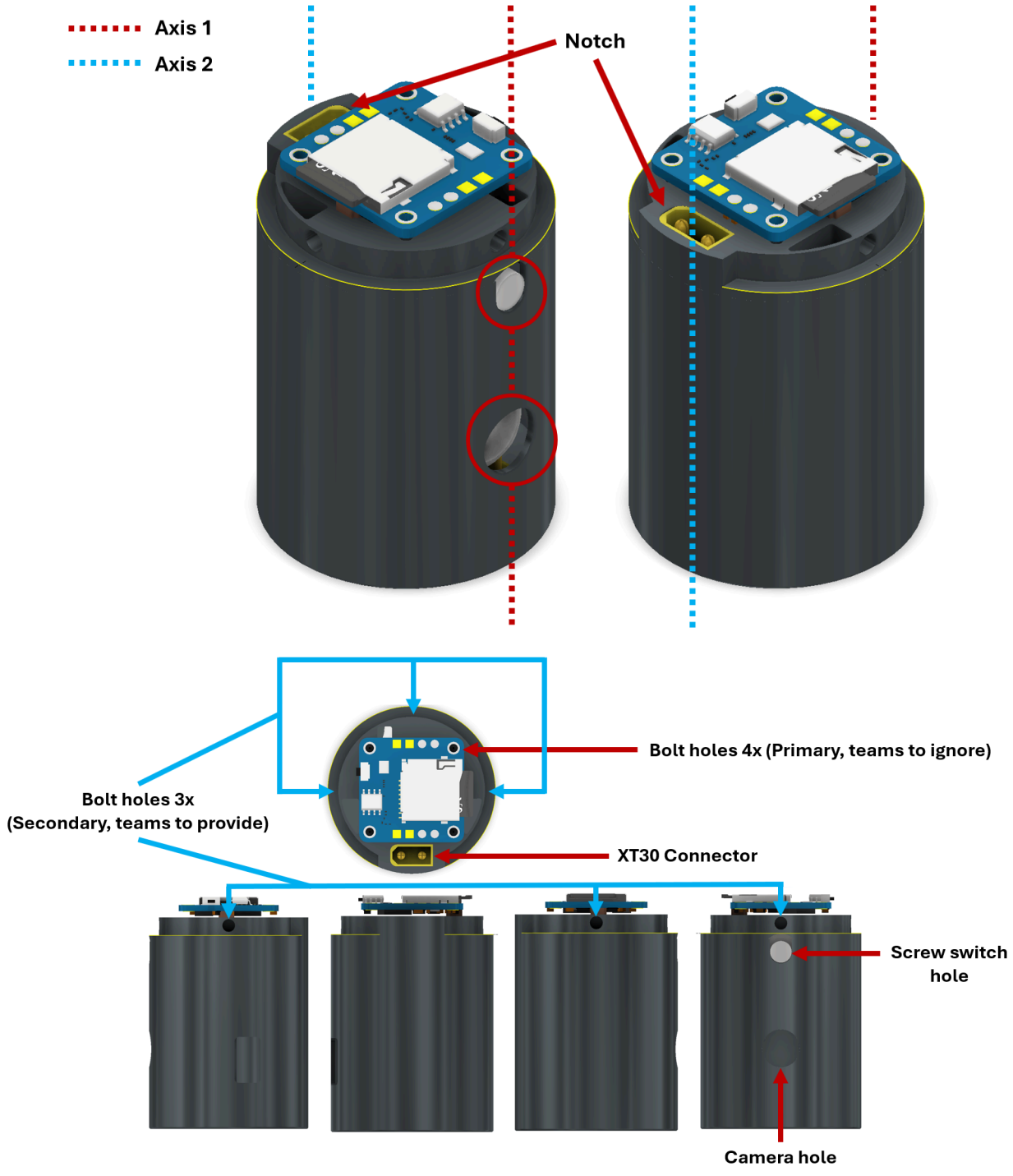


Figure RS-16: Customer Payload Alignment

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VV Method: D

Parent: N/A

NRC RS-MRD-CPD-010 | Rev: A

The launch vehicle payload bay shall be unpressurised.

This allows "The Customer's" payload to read the altitude of its launch vehicle.

VV Method: D

Parent: N/A

NRC MRD-CPD-011 | Rev: A

Teams shall provide three M3 bolts to mount "the Customer's" payload to their launch vehicle adapter.

This is to "mate" the payload adapter and the payload itself. As standard, three M3 bolts are required. Teams are advised to bring spare bolts in a variety of lengths, in case they are dropped in the field at the launch site and are lost in the grass!

UKSEDS will have a small supply of spare M3 bolts in various sizes. We have added this requirement because many previous teams modified their launch vehicle adapter to use bolts of different lengths..

VV Method: D

Parent: NRC RS-MRD-CPD-006, NRC RS-MRD-CPD-009

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11. Rideshare Payload [RPD]

NRC RS-MRD-RPD-001 | Rev: A

The rideshare payload shall contain a printed circuit board designed from scratch.

Teams should design, test and integrate a Printed Circuit Board (PCB) from scratch. UKSEDS have provided a training course on using KiCAD to build your own PCB. This PCB should be designed dedicated for this year's competition (teams may not use a previous year's design), and is bound by the Code of Conduct.

VV Method: *D*

Parent: *N/A*

NRC RS-MRD-RPD-002 | Rev: A

The rideshare payload shall detect the launch vehicle's altitude.

IMU's are inclined to quickly drift over time and return inaccurate results. Therefore barometers are recommended in respect to this requirement.

VV Method: *D*

Parent: NRC RS-MRD-RPD-001

NRC RS-MRD-RPD-003 | Rev: A

The rideshare payload shall be able to produce an altitude vs time plot (in feet against seconds) within 10 minutes of recovery.

Note: the payload *does* not *require* a screen to display this plot. Teams may use an external screen such as a laptop if desired.

VV Method: *D*

Parent: NRC RS-MRD-RPD-001

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NRC RS-MRD-RPD-004 | Rev: A

The rideshare payload shall be able to provide an apogee value in feet within 10 minutes of recovery without the use of external equipment.

Note: mechanical tools (screwdrivers, impact-drivers etc) are allowed to gain access to the rideshare payload. External digital equipment such as laptops to retrieve the data from the rideshare payload are prohibited.

VV Method: *D*

Parent: NRC MRD-RPD-001

NRC RS-MRD-RPD-005 | Rev: A

The rideshare payload shall include a silkscreen on the printed circuit board displaying the team logo.

A silkscreen is a static print on the board, not an electrical component.

VV Method: *D*

Parent: NRC RS-MRD-RPD-001

NRC RS-MRD-RPD-006 | Rev: A

The rideshare payload shall record at least one additional type of data and present it within 10 minutes of recovery.

Note: teams may use external equipment to display this data. This data must not be collected with the sensor used for NRC RS-MRD-RPD-002.

VV Method: *D*

Parent: NRC RS-MRD-RPD-001

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12. Launch Vehicle Readiness [FVR]

The launch vehicle is expected to arrive at the MRC in a near flight-ready condition. Launch day activities are intended for final assembly, payload integration, inspection, and other minor preparation tasks, rather than for completing manufacture, development, or unresolved technical work.

As a large number of teams must be processed within a limited launch-day schedule, these requirements are intended to minimise delays, support an efficient inspection process, and maximise the likelihood that all teams are ready to launch at their allocated times.

NRC RS-MRD-FVR-001 | Rev: A

All launch vehicle subsystems required for flight shall be fully manufactured, integrated to the extent practicable, and functionally tested prior to arrival at the MRC on launch day.

VV Method: I

Parent: N/A

NRC RS-MRD-FVR-002 | Rev: A

All known technical issues that may affect flight safety, recovery system performance, vehicle stability, structural integrity, propulsion, avionics, or other flight-critical functions shall be resolved and, where applicable, reverified prior to arrival at the MRC on launch day.

VV Method: I

Parent: N/A

NRC RS-MRD-FVR-003 | Rev: A

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Work performed on the launch vehicle at the MRC on launch day shall be limited to final assembly, customer payload integration, installation of flight consumables, repairs, and other minor flight preparation activities.

Activities that constitute manufacturing, or continued development of the vehicle are not allowed.

VV Method: D

Parent: N/A

NRC RS-MRD-FVR-004 | Rev: A

The total planned duration of launch vehicle assembly and preparation activities performed on launch day shall not exceed 45 minutes

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This is excluding activities performed as part of official inspection, range operations, or actions specifically requested by UKSEDS personnel.

VV Method: D

Parent: N/A

NRC RS-MRD-FVR-005 | Rev: A

Launch-day work on the vehicle shall only require basic hand tools suitable for final assembly and integration, such as screwdrivers, spanners, wrenches, hex keys, and similar non-powered tools.

Activities requiring machining, drilling, cutting, soldering, grinding, composite manufacture, adhesive curing, or other fabrication processes should not be a part of the planned launch-day preparation.

VV Method: D

Parent: N/A

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13. Further Design Considerations

The previous sections outline the minimum requirements intended to help teams maximise the performance of their launch vehicles while minimising associated risks and safety hazards. There are, however, additional design considerations that are equally important but are difficult to define through fixed requirements, as they depend heavily on the specific design and configuration of each launch vehicle.

This section provides a brief overview of these considerations and is intended to guide teams when making design, integration, and manufacturing decisions throughout the development of their launch vehicle.

13.1. Fins Design

Fin thickness and cross-sectional geometry influence both aerodynamic performance and structural behaviour. Thinner and appropriately profiled fins can reduce aerodynamic drag, particularly at higher flight velocities. However, reducing fin thickness also decreases structural stiffness and can increase fin deflection and susceptibility to aeroelastic effects such as flutter. Teams should therefore avoid selecting fin thickness based solely on drag reduction. The expected flight velocity, Mach number, fin span, vehicle mass, motor thrust profile, aerodynamic loading, and material properties should all be considered when determining an appropriate fin thickness and profile.

The fin profile itself should also be considered. Leading-edge shape, trailing-edge geometry, and the overall thickness distribution can influence drag and aerodynamic loading, particularly as the vehicle approaches transonic or supersonic conditions. Teams operating in these regimes should consider the effects of compressibility and wave drag as part of their aerodynamic analysis.

Where appropriate, teams should also demonstrate that the selected fin design provides sufficient margin against aeroelastic flutter throughout the expected flight envelope.

13.1.1. Fin Thickness and Cross-Section

Fin thickness and cross-sectional geometry influence both aerodynamic performance and structural behaviour. Thinner and appropriately profiled fins can reduce aerodynamic drag, particularly at higher flight velocities, but reducing thickness also decreases stiffness and can increase fin deflection and susceptibility to aeroelastic effects such as flutter. Teams should therefore avoid selecting fin thickness based solely on drag reduction and instead consider the expected flight velocity, Mach number, fin span, vehicle mass, motor thrust profile, and resulting aerodynamic loads.

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13.1.2. Choice of Material

Material selection is closely linked to the fin geometry and expected flight loads. Suitable materials may include plywood, aluminium alloys, glass-fibre composites, carbon-fibre composites, and other engineered materials, depending on the application. Each material presents different trade-offs in strength, stiffness, density, manufacturability, durability, and cost. Teams should consider both strength and stiffness when evaluating candidate materials, as a material may be sufficiently strong to avoid structural failure while still being too flexible for the intended application.

The selected material should ultimately be appropriate for the expected aerodynamic loads and flight conditions while remaining compatible with the chosen fin geometry, manufacturing method, and attachment system.

13.2. Motor Ejection Charge

Some rocket motors incorporate an ejection charge that can be used to initiate recovery system deployment after a predetermined delay following motor burnout. This provides a relatively simple and passive deployment method, as it does not rely on onboard sensing or active flight-state detection.

However, because deployment occurs after a fixed delay rather than in response to the vehicle reaching apogee, the selected delay must closely match the expected flight profile. If the delay is too short, the recovery system may deploy while the vehicle is still ascending. If it is too long, deployment may occur after the vehicle has passed apogee and begun descending. Both cases can impose significant loads on the recovery system and launch vehicle. Some motors, especially single-use motors, provide only a limited number of discrete delay settings, typically adjusted from a maximum delay value. For example, a motor may offer delay options such as 13, 10, 8, 6, or 4 seconds. This can limit how closely the deployment timing can be matched to the predicted time to apogee.

Teams should carefully review the available delay options for their selected motor and compare them against the predicted time to apogee from their OpenRocket simulations. Appropriate margin should also be considered for variations between simulated and actual flight performance.

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